TEACHING GUIDE

Network Demos

Big, looping animations for whole-class discussion. Put one on the projector and talk through it — each demo loops automatically with a plain-English caption for every step.

Using the demos in class

- Pick a topic from the menu bar. The demo starts playing straight away and **loops**.
- **Pause** (⏸) to stop on a step and discuss it; **Play** (▶) to resume.
- ‹ › step backward/forward one phase at a time — handy for questioning.
- ↻ restarts from the beginning; the **speed** control slows things down or speeds them up.
- Press **F11** for full-screen on the projector. Everything scales to fit.

What each demo shows

Demo	Key idea to draw out
OSI encapsulation	Headers are added going down (Data→Segment→Packet→Frame→Bits) and stripped going up.
Routing between networks	IP addresses are end-to-end; MAC addresses change at every hop.
VLANs & trunking	VLANs are separate broadcast domains; trunks tag frames (802.1Q) to carry several VLANs.
OSPF convergence	Hello → LSA flooding → SPF (shortest cost) → routing table; it re-routes after a failure.
NAT / PAT	Private IPs are rewritten to one public IP; a port number keeps each conversation apart.
TCP handshake	SYN → SYN-ACK → ACK synchronises sequence numbers before any data flows.
DHCP (DORA)	Discover, Offer, Request, Acknowledge — how a client gets its address.
ARP	Broadcast "who has this IP?" → unicast reply → cache the MAC in the ARP table.
Subnetting	Borrowing host bits doubles the number of subnets and halves the hosts in each.

Discussion prompts

- "Why didn't the IP address change as the packet crossed the routers?"
- "What would happen to the VLAN frame if the trunk wasn't configured?"
- "After the OSPF link fails, why is the new path's cost higher?"
- "With PAT, how does the router know which inside host a reply belongs to?"

Tip: pause on a step and ask the class to predict the next one before you press play.